

Multiple Choice (10 marks)

1. Which of the following is true of the assessment of power in the arms?
 - (a) Extension at the elbow is produced by the biceps muscle
 - (b) Grade 3/5 (MRC scale) represents movement of the arm only when gravity is eliminated
 - (c) Grade 2/5 power represents movement of the arm only against gravity.
 - (d) The brachioradialis muscle is responsible for shoulder abduction
 - (e) Grade 0/5 power represents full range of motion with resistance
2. The influence of genetics on longevity is estimated to be about
 - (a) 33%
 - (b) 40%
 - (c) 60%
 - (d) 20%
 - (e) 80%
3. The World Health Organization recommends zinc supplements for which groups of people?
 - (a) Men with high physical activity levels
 - (b) All women over the age of 50
 - (c) All children in high income areas with low prevalence of stunting
 - (d) Elderly people with low incomes
 - (e) Adolescents in high-stress environments
4. The coronal suture joins the
 - (a) parietal, squamous temporal and greater wing of the sphenoid.
 - (b) left and right parietal bones.
 - (c) parietal and sphenoid bones.
 - (d) frontal and occipital bones.
 - (e) squamous temporal and greater wing of the sphenoid.
5. Which of the following appears to lower bad cholesterol?
 - (a) Vitamin E
 - (b) Vitamin A

- (c) Thiamine
- (d) Niacin
- (e) Vitamin B12

True / False (10 marks)

Answer True or False

6. True or False: Parvoviruses only replicate in dividing cells, which limits their ability to infect non-dividing tissues.
7. True or False: Among the various factors studied, behavioral factors have received significant research support for their link to social support and health.
8. True or False: Enzyme assays are primarily used to diagnose carriers of Huntington's disease based on genetic mutations.
9. True or False: People with no previous history of mental health issues are more likely to experience depression after a loss.
10. True or False: The cardiac cycle includes the phases of systole, diastole, and rest, which are critical for blood circulation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a relative risk ratio of 1.2 for hip fractures in elderly women who exercise regularly, considering the confidence interval and its significance in health outcomes.
12. Discuss the implications of a mutation in the repressor of the lac operon in *E. coli* that impedes its binding to lactose, particularly in the context of low glucose levels and gene expression.

End of Paper